PART 2: CHAPTER 2.3 — DATA SECURITY & ENCRYPTION (TB Pages 63–74)

When data is transmitted across public networks (such as the Internet), unauthorized eavesdroppers can intercept packets. **Encryption** transforms confidential data into an unreadable scrambled form to safeguard privacy.

1. Core Terminology — TB Page 63

- **Plaintext:** The original, readable source data before encryption is applied (e.g., `"Meet at 10 AM"`).
- **Ciphertext:** The scrambled, unreadable output produced after running plaintext through an encryption algorithm using an encryption key.
- **Encryption Algorithm:** The mathematical formula used to scramble the plaintext.
- **Key:** A binary sequence or secret value required by the algorithm to encrypt and decrypt the message.

2. Symmetric Encryption — TB Pages 63–67

In **Symmetric Encryption**, the exact same single secret key is used to both *encrypt* the plaintext and *decrypt* the ciphertext.

```
[Plaintext] + [Shared Secret Key] → [Ciphertext]
[Ciphertext] + [Shared Secret Key] → [Plaintext]
```

- **Advantage:** Computationally fast and efficient; ideal for encrypting massive files, hard drives, and streaming video.
- **The Key Distribution Problem:** Both the sender and recipient must possess the identical secret key. How do they share this key initially across an insecure network? If a hacker intercepts the key in transit, all future messages can be easily decrypted!
- **Brute Force Attacks:** Attackers use high-speed automated supercomputers to test every possible key combination until the correct one is discovered. Longer keys (e.g. AES 256-bit) make brute-force attacks mathematically impossible within human lifespans.

3. Asymmetric Encryption — TB Pages 68–70

To permanently solve the Key Distribution Problem, **Asymmetric Encryption** utilizes a mathematically linked pair of different keys:

1. **Public Key:** Made freely available to anyone in the world (e.g. published on a website directory). Anyone can use it to encrypt a message intended for the owner.
2. **Private Key:** Kept strictly confidential and secret by the owner. It is NEVER shared or transmitted across any network.

```
Rule 1: What is encrypted with the PUBLIC key can ONLY be decrypted by the
matching PRIVATE key.
Rule 2: It is mathematically infeasible to deduce the private key from the public
key.
```